$$\partial_p(\sigma) = \partial_p[x_1, x_2, \ldots, x_p] = \sum_{i=0}^{p} [x_1, \ldots, \hat{x}_i, \ldots, x_p].$$

It can be verified that for any p-simplex σ, $p \geq 1$, we have $\partial_{p-1}\partial_p(\sigma) = 0$. This property of boundary operators is also referred as ***fundamental lemma of simplicial homology****, which is essential to define homology. Here, it should be noted that by convention, the boundary operator ∂_0 maps a 0-simplex to an empty chain 0. Besides, the fundamental lemma of simplicial homology allows us to define an algebraic structure, called chain complex which allows us to compute homology of K.*

Definition 2.17 ***Chain complex:*** *The chain complex of a finite simplicial complex K of dimension n is the sequence of vector spaces $C_n(K, \mathbb{F})$ connected with the corresponding boundary operators:*

$$0 \xrightarrow{\partial_{n+1}} C_n(K, \mathbb{F}) \xrightarrow{\partial_n} C_{n-1}(K, \mathbb{F}) \xrightarrow{\partial_{n-1}} \ldots \xrightarrow{\partial_2} C_1(K, \mathbb{F}) \xrightarrow{\partial_1} C_0(K, \mathbb{F}) \xrightarrow{\partial_0} 0.$$

For, $p \in \{0, \ldots n\}$, we define the set of p-cycles as follows:

$$Z_p(K) := \text{ kernel } (\partial_p : C_p(K, \mathbb{F}) \to C_{p-1}(K, \mathbb{F})) = \{\gamma \in C_p(K, \mathbb{F}) : \partial_p(\gamma) = 0\}.$$

The set of p-boundaries is defined as follows:

$$\begin{aligned} B_p(K) &:= \textit{image}(\partial_{p+1} : C_{p+1}(K, \mathbb{F}) \to C_p(K, \mathbb{F})) \\ &= \{\gamma \in C_p(K, \mathbb{F}) : \gamma = \partial_{p+1}(\gamma^*), \text{ for some } \gamma^* \in C_{p+1}(K, \mathbb{F})\} \end{aligned}$$

Definition 2.18 ***Homology:*** *Given a finite simplicial complex K and its associated chain complex, it is evident that $B_p(K)$ and $Z_p(K)$ are subspaces of $C_p(K, \mathbb{F})$ and according to fundamental lemma of simplicial homology, we have the following:*

$$B_p(K) \subset Z_p(K) \subset C_p(K, \mathbb{F}).$$

For any non-negative integer p, we denote the pth homology of K as $H_p(K)$, which is defined to be the following quotient vector space:

$$H_p(K) := Z_p(K)/B_p(K)$$

Note that $H_p(K)$ is a vector space and its elements are called homology classes of K.

Definition 2.19 ***Betti numbers:*** *The pth Betti number β_p is defined to be the dimension of the vector space $H_p(K)$, i.e $\beta_p :=$ dim $(H_p(K))$.*

Note that the calculation of Betti numbers involves choosing a value $\epsilon > 0$ to construct a simplicial complex K, which makes Betti numbers unstable. Suppose that K_ϵ denotes a simplicial complex constructed at a fixed value of ϵ, then note that the homology of K_ϵ changes as ϵ changes. Therefore, instead of choosing a single value of ϵ, one tracks the evolution of topological features over all the possible values of ϵ. This idea leads to the notion of persistent homology introduced by Edelsbrunner und Harer (2008).

Definition 2.20 ***Filtrations:*** *A filtration of a simplicial complex K is a nested family of sub-complexes $(K_\epsilon)_{\epsilon \in T}$, where $T \subset \mathbb{R}$, such that for any $\epsilon, \epsilon^{'} \in T$ we have the following:*

- $K = \bigcup_{\epsilon \in T} K_\epsilon$.
- $K_\epsilon \subseteq K_{\epsilon'}$, *for any* $\epsilon \leq \epsilon^{'} \in T$.

Definition 2.21 ***Persistent homology:*** *For a given filtration $(K_\epsilon)_{\epsilon \in T}$ of a simplicial complex K and for a non-negative integer p, we obtain a sequence of pth-homology vector spaces $H_p(K_\epsilon)$, where the inclusions $K_\epsilon \subseteq K_{\epsilon'}$, for any $\epsilon \leq \epsilon^{'} \in T$, induce linear maps $f_p^{\epsilon,\epsilon^{'}}$ between $H_p(K_\epsilon)$ and $H_p(K_{\epsilon'})$. Such a sequence of vector spaces together with the linear maps connecting them is called the persistence module. Therefore, given a persistence module associated with the simplicial complex K, we define the pth persistent homology $H_p^{\epsilon,\epsilon^{'}}(K)$ as follows:*

$$H_p^{\epsilon,\epsilon^{'}}(K) := Z_p(K_\epsilon)/(B_p(K_\epsilon) \cap Z_p(K_{\epsilon'})),$$

where $Z_p(K)$ and $B_p(K)$ denote the set of p-cycles and the set of p-boundaries of a simplicial complex K, respectively. Note that here, $H_p^{\epsilon,\epsilon^{'}}(K)$ contains p-dimensional holes of K_ϵ that are still present in $K_{\epsilon'}$. Also, note that $H_p^{\epsilon,\epsilon}(K) = H_p(K_\epsilon)$.

Definition 2.22 ***Persistent Betti numbers:*** *For any $\epsilon \leq \epsilon^{'}$, the pth persistent Betti number $\beta_p^{\epsilon,\epsilon^{'}}$ is defined to be the dimension of the vector space $H_p^{\epsilon,\epsilon^{'}}(K)$, i.e, $\beta_p^{\epsilon,\epsilon^{'}} :=$ dim $(H_p^{\epsilon,\epsilon^{'}}(K))$.*

Note that, for $\epsilon = \epsilon^{'}$, the pth persistent Betti number $\beta_p^{\epsilon,\epsilon^{'}}$ coincide with the usual Betti number of K_ϵ.

3 Problem Formulation and Main Results

We are given a random sample $X_1, \ldots, X_n$ from some probability measure $\mathbb{P}$ supported on an unknown compact set $\mathcal{X} \subseteq \mathbb{R}^d$, $d \geq 1$, and our goal is to infer the topology of the unknown set $\mathcal{X}$, based on the observed sample. In particular, we are interested in the homology of the set $\mathcal{X}$ based on the observed data. In this study, we assume $\mathcal{X}$ to be a compact manifold (See, Definition 2.5) embedded in $\mathbb{R}^d, d \geq 1$.

Our goal is to perform one sample test for homological equivalence of $\mathcal{X}$ to some hypothesized set $\mathcal{X}_0$. Moreover, we shall extend this setup to two sample problems. As Betti numbers characterize the homology of a topological space, we want to test the statistical significance of calculated Betti numbers of $\mathcal{X}$ to the Betti numbers of $\mathcal{X}_0$. Further, since $\mathcal{X}$ is known only up to the random sample $X_1, X_2, \ldots, X_n$, we approximate the topological space $\mathcal{X}$ by a simplicial complex constructed from the random sample $X_1, X_2, \ldots, X_n$. We consider Čech and Vietoris-Rips complex as a simplicial complex constructed from the data to approximate the topological space $\mathcal{X}$. Specifically, Čech complex is used for theoretical purposes, and the Vietoris-Rips complex is used for computational purposes to calculate the Betti numbers.

Note that, one needs to choose a positive threshold ϵ to construct Čech or Rips complex, the threshold ϵ, also referred to as 'proximity parameter'. The threshold ϵ is a tuning parameter, and one can view topological features at different scales by changing the value of ϵ. Therefore, the behavior of the Čech and Rips complex depends on the threshold ϵ. Since we are interested in the asymptotic properties of these random complexes, we choose ϵ as a function of sample size n, i.e., $\epsilon \equiv \epsilon(n)$ such that $\epsilon(n) \to 0$ as $n \to \infty$. Thus, depending on $\epsilon(n)$, there are three regimes in which the limiting behavior of the random complexes differs significantly. The subcritical regime is when $n\epsilon^d \to 0$ as $n \to \infty$, the critical regime (the thermodynamic regime) is when $n\epsilon^d \to \theta \in (0, \infty)$ as $n \to \infty$ and the supercritical regime is when $n\epsilon^d \to \infty$ as $n \to \infty$, $d \geq 1$ is dimension of the data. We establish consistency of the proposed tests under the critical